A Comparative Analysis of Weapon System Design in *PUBG* and *Apex Legends*

Apex Legends and *PUBG: Battlegrounds*, as two "pillars" of the battle royale genre, provide significantly different—and in some aspects, completely opposite—player experiences:

- *Apex Legends* emphasizes fast-paced, mechanically driven hero-based combat with a strong focus on fluid shooting;
- *PUBG: Battlegrounds* emphasizes information asymmetry, tactical decision-making, and a high-stakes "one-shot, one-kill" survival simulation.

Differences across various design elements lead to distinct gameplay styles. Below, this analysis focuses on the **weapon system** to examine the macro-level differences, design motivations, and the unique experiences each game delivers.

- **1.1 Macro Differences in Weapon Systems**

The weapon systems in *Apex Legends* and *PUBG: Battlegrounds* represent two fundamentally different design philosophies:

- In *Apex Legends*, weapons are generally easier to handle, serving the game's fast-paced shooting experience;
- In *PUBG*, weapons are more "unstable," emphasizing realism and a higher skill entry barrier.

We analyze these differences from two perspectives:

- **Mechanics layer** (bullet trajectory, handling, weapon roles, etc.)
- **Numerical layer** (DPS, TTK, etc.)

- **1.1.1 Differences at the Mechanics Layer**

When discussing shooting mechanics, the most intuitive aspects are **bullet trajectory** and **recoil**.

In *PUBG*, bullets exhibit noticeable drop and travel time. In contrast, *Apex Legends* features faster projectile speeds and reduced drop, resulting in a more "point-and-shoot" feel.

Additionally, players can clearly feel that weapon handling in *PUBG* is more difficult, with a higher skill ceiling and learning cost compared to *Apex Legends*. The differences can be summarized as follows:

Category	PUBG	Apex Legends
Hit Logic	Weapon-driven, with strong physical simulation	Instant-hit feel, gameplay-first

Control	Strong recoil, random spread patterns	Lower recoil, more predictable patterns
Attachments	Highly impactful	Less impactful
Weapon Role	Strong range specialization, primary combat variable	Primarily a damage-dealing tool synergizing with abilities

It is evident that *Apex Legends* has a significantly lower barrier to entry in terms of weapon handling—one of the most important distinctions in its design philosophy.

In *Apex Legends*, weapons are tools for output rather than a core skill barrier. This shifts the player skill emphasis toward movement, tracking, and mechanical execution, supporting the game's design goal of **"sustained combat + high skill ceiling."** It is also a major source of its shooting satisfaction.

In contrast, *PUBG* uses high weapon handling difficulty as a core skill test. Different weapons come with varying levels of difficulty, increasing the learning cost. The weapon system supports the game's **"survival simulation + tactical decision-making"** philosophy, aiming to recreate a more realistic battlefield while enforcing meaningful weapon selection and loadout constraints.

- **1.1.2 Differences at the Numerical Layer**

Next, we compare the two games from a numerical perspective, focusing on **DPS** and **TTK**.

- **DPS (Damage Per Second)**
- In *Apex Legends*, DPS varies across weapon types and individual weapons, but does not create a strict hierarchy of "high-tier vs. low-tier" weapons. Instead, weapons are designed to be **interchangeable based on player preference**, with playstyle and engagement distance being the primary factors in weapon choice.
- In *PUBG*, there are more pronounced differences in DPS and single-shot damage. Weapons that are harder to control often offer higher DPS and burst potential, resulting in faster kill windows. This makes player weapon proficiency a more critical factor in decision-making.

These differences serve each game's design goals:

- *Apex Legends* allows players to choose weapons that fit their preferred playstyle, minimizing strict weapon tier gaps while still loosely tying DPS to handling difficulty.
- *PUBG* treats weapons as a core skill check, with stronger weapons demanding higher mastery. This increases the learning curve and aligns with its realism-driven survival design.

- **TTK (Time to Kill)**

From a TTK perspective:

- *Apex Legends* features significantly longer TTK compared to *PUBG*, providing players with a larger reaction window and higher tolerance for mistakes.

This is a fundamental difference in core experience, reflecting:

- *Apex Legends*: sustained engagements and continuous combat
- *PUBG*: high lethality and "one-shot determines survival"